MILITARY SPECIFICATION

TEMPERATURE MEASUREMENT EQUIPMENT

SIGNAL CONDITIONER AND POWER SUPPLY

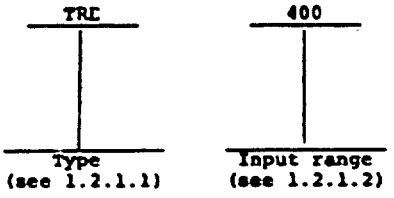
(ELECTRICAL) (NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope. This specification covers the requirements for signal conditioners and power supplies (electrical) used in conjunction with thermocouples and resistance temperature element assemblies for Naval ships. It does not include the design of the sensing elements and wells or the requirements for the readout or display.

1.2 Classification.

1.2.1 The classification shall be of the following format, as specified (see 6.2.1):



1.2.1.1 Type. Temperature measurement equipment shall be designated by the three letter symbols as follows:

TRE - Temperature resistance equipment, platinum.
TTE - Temperature thermocouple equipment, type K.

1.2.1.2 Input range. The temperature range in degrees Fahrenheit (°F) shall be designated by its numerical value (see 3.5.1.1 and 3.5.2.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-M-15071 - Manuals, Technical, Equipment and Systems, Content Requirements for.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-S-19500 - Semiconductor Devices, General Specification for.

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MILITARY (Cont'd)

MIL-J-24142 - Junction Boxes for Electrical Fittings and Fixtures, General Specification for.
MIL-J-24142/3 - Junction Box, Submersible, Size 6 by 9.
MIL-T-24388 - Thermocouples and Resistance Temperature Element Assemblies (Naval Shipboard Use).
MIL-P-24423 - Propulsion and Auxiliary Control Consoles and Associated Control and Instrumentation Equipment, Naval Shipboard Use, Basic Design Requirements.
MIL-I-45208 - Inspection System Requirements.
MIL-P-55110 - Printed Wiring Boards.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types, and Associated Accessories, General Specification for.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-758 - Packaging Procedures for Submarine Repair Parts Utilizing Transparent, Flexible, Heat Sealable Film.
MIL-STD-831 - Test Reports, Preparation of.

HANDBOOK

MIL-HDBK-H-1 - Cross-Index of Chemical by Equivalent Specifications and Identification Code (Ferrous and Nonferrous alloys).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Y32.16-68 - Reference Designations for Electrical and Electronic Parts and Equipments.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

INSTRUMENT SOCIETY OF AMERICA

Recommended Practice 37.1 - Nomenclature and Specification Terminology for Aerospace Test Transducers with Electrical Output.

(Application for copies should be addressed to the Instrument Society of America, Penn Sheraton Hotel, 530 William Penn Place, Pittsburgh, Pa. 15219.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The temperature measurement equipment furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.4).

3.2 Definitions. Terminology consistent with Instrument Society of America Recommended Practice 37.1 shall apply, except as modified by the definitions listed as follows:

(a) Thermocouple. Shall be as defined by MIL-T-24388.
(b) Resistance temperature element. Shall be as defined by MIL-T-24388.
(c) Deadband. The range through which the measurand can be verified without a change in output.

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(d) Static error band. The maximum deviation from a straight line drawn through the coordinates of the lower span limit at specified output, and the upper span limit at specified output expressed in percent of span.

3.3 Description. The equipment specified herein in conjunction with the thermocouples or resistance temperature elements specified in MIL-T-24438 comprise a temperature instrument. The temperature measurement equipment shall consist of the following units:

(a) Signal conditioner. The signal conditioner shall convert the sensing element output to a continuous linear analog signal directly proportional to temperature.
(b) Power supply. The power supply shall provide excitation energy to the signal conditioner and sensor.
(c) Test device. A test device shall be furnished to provide a calibrated test signal used for calibrating the equipment.

3.3.1 The assemblies specified in 3.3 shall be built integrally together and housed in the same enclosure.

3.4 General features. The temperature measurement equipment shall be in accordance with section 3 of MIL-P-24423 and as specified herein. (Whenever a requirement of MIL-P-24423 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

3.4.1 Materials. Except where specifications are referenced, materials shall be in accordance with commercial specifications having material compositions considered suitable for the intended service. Where commercial specifications are referenced, equivalent materials shall be in accordance with MIL-HDBK-H-1.

3.4.2 Temperature measurement equipment assembly - general design requirements.

3.4.2.1 Equipment shall be designed for an expected life of 40,000 hours of operation when installed and subjected to the following conditions encountered in Naval service:

(a) High shock (see 4.8.14).
(b) Vibration (see 4.8.13).
(c) Temperature (see 4.8.11), temperature measurement equipment shall have an ambient operating temperature range: 0 to 65 Celsius (C).
(d) Supply voltage and frequency variations (see 4.8.8 and 4.8.10).
(e) Inclination (see 4.8.6).

3.4.2.2 Output. The electrical signal output of the equipment shall be dc, directly proportional to temperature input. The output shall be 4 to 20 ma, dc into an external resistance of 550 ± 10 percent ohms.

3.4.2.3 Calibration means. A means shall be provided to monitor equipment output corresponding to temperature input, to permit in place calibration by one man. Each temperature instrument shall contain a test switch and a temperature detector simulator, capable of supplying a test signal into the instrument circuitry. This test device shall have a calibrated dial and shall replace the detector when the instrument is in the test mode. The signal from this simulator shall be of sufficient accuracy to determine static error band (see 3.10.1) and repeatability (see 3.10.2). Test jacks shall be provided to monitor the equipment output and detector output. Placing the instrument in the test mode shall allow remote monitoring of the detector. Calibration shall be effected without the necessity of electrical disconnection.

3.4.2.4 Equipment range. Equipment range shall be determined by means of an interchangeable assembly or internal adjustments.

3.4.2.5 Adjustments. Tamper-proof adjustments for zero and span shall be provided for calibration purposes. The number of adjustments shall be kept to a minimum consistent with the operation and maintenance of the equipment and the elimination for the selective matching of parts.

3.4.2.6 Failsafe output. If the input of the signal conditioner is open circuited, the output shall drive upscale or downscale, the choice of which shall be made by means of a link.

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3.4.2.7 Isolation. Input and output circuits shall be isolated from each other and from ground.

3.4.2.8 Enclosure. Temperature measurement equipment shall be mounted in a junction box as specified in MIL-J-24142 and MIL-J-24142/3. All adjustments and test points shall be accessible when the cover is removed.

3.4.2.9 Enclosure mounting. Temperature measurement equipment shall be designed for bulkhead or bracket mounting (see 6.2.1).

3.4.2.10 Weight. The weight of the complete equipment shall be kept to a minimum commensurate with good engineering design.

3.4.2.11 Power supply requirements. The equipment shall be designed to operate with line variations as specified in 4.8.8 and 4.8.10. Nominal steady state power supply requirements shall be 115 ± 10 percent volts at 60 ± 5 percent hertz (Hz), single phase, unless otherwise specified (see 6.2.1).

3.4.2.12 Cable entrance and connection. Cable entrance shall be by means of stuffing tubes located on the bottom surface of the enclosure.

3.4.2.13 Operation. The operation of the equipment shall not depend on balancing motor, slidewires or similar devices. The system shall be completely static. The use of potentiometers are for test and adjustment purposes only.

3.5 Type requirements.

3.5.1 Temperature resistance input.

3.5.1.1 Ranges. Equipment temperature range shall be as specified (see 6.2.1). Ranges shall be in accordance with the following (see 3.5.1.2):

-40° to 260°F
0° to 400°F
0° to 1000°F

3.5.1.2 Resistance temperature element shall be platinum conforming to the temperature resistance characteristics of MIL-T-24388.

3.5.1.3 Excitation current. The maximum current through the resistance temperature element shall be 6 ma, dc.

3.5.1.4 Lead wire resistance compensation. The equipment input for resistance sensing technique shall be of three wire configuration. A means shall be provided to compensate for self heating error and the resistance of interconnecting wiring between the sensor and signal conditioner.

3.5.2 Temperature thermocouple input.

3.5.2.1 Ranges. Equipment temperature range shall be as specified (see 6.2.1). Ranges shall be in accordance with the following (see 3.5.2.2):

-40° to 200°F
0° to 400°F
0° to 1000°F
400° to 1200°F
500° to 1500°F

3.5.2.2 Thermocouple shall be type TCE in accordance with MIL-T-24388.

3.5.2.3 Cold junction compensation. Cold junction compensation for thermocouple sensing technique shall be automatic and shall be referenced to 32°F.

3.6 Detail requirements.

3.6.1 Semiconductors. Silicon rectifiers, diodes, transistors and other semiconductor devices shall be in accordance with MIL-S-19500 as applicable and shall be selected from MIL-STD-701. Devices other than those listed in MIL-STD-701 shall be used only where

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specifically approved in writing by the Naval Ship Engineering Center (NAVSEC) (Submarine and Advanced Concepts Branch) on the basis of justification submitted in accordance with requirements of MIL-STD-701. Any such device shall comply with an existing Military specification (device detail specification); or, if none exists, the requirements of 3.6.1.1 shall apply.

3.6.1.1 Semiconductor device special specifications. If approval is sought, in accordance with 3.6.1, for a device for which there is no existing detail specification issued under MIL-S-19500, or in other acceptable form, a special detail specification shall be submitted by the contractor with the request for approval. The special specification shall, insofar as practicable, use the detail specification for the MIL-S-19500 device of comparable construction which is closest in size to the device for which approval is sought. The MIL-S-19500 detail specification so used is referred to herein as the guide specification. Insofar as they are appropriate, requirements of the guide specification shall be invoked by the special specification, and the device manufacturer's latest published commercial specification rating and characteristic values shall be substituted for any inapplicable values in the guide specification to form the new special specifications. The changes in, and additions to, the guide specification as set forth in the special specification shall be in sufficient detail and completeness to enable direct Government procurement of completely acceptable devices from any device manufacturer who has the production capability and willingness to comply with the special specification. The special specification shall be assigned a drawing number, and, when approved, shall form a part of the set of drawings covering the equipment.

3.6.2 Electron tubes and vibrators. Electron tubes and vibrators shall not be used.

3.6.3 Resistors. Composition type fixed or variable resistors shall not be used.

3.6.4 Batteries. Batteries shall not be used.

3.6.5 Modular assemblies. The configuration of the temperature measurement equipment shall meet the modular construction requirements specified in 3.6.5.1 through 3.6.5.3.

3.6.5.1 Module dimensions, packaging configuration, potting and encapsulation shall be subject to development by the supplier and acceptable by NAVSEC.

3.6.5.2 The plug-in module units may include, but are not limited to, printed circuit boards for conventional components, cable connectors, power supplies, and integrated microelectronic circuits.

3.6.5.3 Accessibility. Modular units or assemblies shall be fastened in such a manner to allow for quick and easy removal for maintenance accessibility or replacement.

3.6.6 Printed wiring boards. Printed wiring boards shall be designed and constructed in accordance with MIL-STD-275 and MIL-P-55110. Conformal coating shall be type UR.

3.6.7 Terminal boards. The interface with shipboard wiring shall be by means of terminal boards in accordance with MIL-T-55164. These terminals shall be accessible with the coverplate removed.

3.7 Drawings.

3.7.1 General requirements. Drawings shall conform to categories A, B and C, form 2 of MIL-D-1000 and to the detail requirements herein. The number of drawings and the amount of detail shall be held to a minimum consistent with the fulfillment of their function.

3.7.2 Copies of drawings. Unless otherwise specified in the contract or order (see 6.2.1), two copies of the drawings shall be submitted for initial approval to NAVSEC.

3.7.3 Approval of drawings. Drawings shall be approved by NAVSEC. Approval shall be required only on the first submittal of an item and whenever changes are made. Where drawing approval has previously been obtained, it shall be necessary only to submit copies of the approved drawing to meet drawing requirements on subsequent contracts. No changes shall be made without prior approval of NAVSEC.

3.7.4 Engineering drawings. Engineering drawings shall be type II of MIL-D-1000/2 and shall include the following minimum information as applicable:

(a) Descriptive data for the equipment.
(1) Type (see 1.2.1.1).
(2) Input range (see 1.2.1.2).

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(3) Current output (see 3.4.2.2).
(4) A statement that the equipment is in accordance with the requirements of this specification and a list of exceptions to this specification and to referenced specifications.
(5) A list of referenced drawings.
(6) Description of the nature and purpose of any adjustments.
(7) Test procedure data, including letter, report numbers and dates.
(8) Shock and vibration classification.
(9) Enclosure, degree of.
(10) Ambient temperature.
(11) Weight of equipment.
(12) Any special features.

(b) List of material, presenting following data in tabular form:
(1) Piece number (for flags of assembly drawing).
(2) Name of part (for example, transformer, diode, and other similar parts).
(3) Schematic diagram designation for electrical parts (for example, switch S-1, and other similar designations).
(4) Number of each part required per assembly.
(5) Military or Federal Material specification number or NAVSEC drawing number, if any.
(6) Type, class, grade, size, military designation, or other classification of any referenced specification.
(7) Part number or identification assigned by manufacturer.
(8) Name of manufacturer of part.
(9) Part number or identification assigned by part manufacturer.
(10) Rating of part from part manufacturer's catalog (for example, resistor 10 ohm 25 watt; diode - 200 PRV, 750 milliamps, and similar ratings).
(11) Remarks column.

(c) Miscellaneous part and component requirements and application data. Information on parts and components shall include the data specified below. The data required shall be presented in tabular form whenever practical. Each part shall be identified by its reference designation:
(1) Transformer, reactor and inductor data:
a. Core data:
1. Lamination form - EI, toroid, etc.
2. Lamination material and gage.
3. Stack and interleave dimensions.
4. Construction - open, tape wound, cut core, and other construction.
5. Core box, if any.
6. Ground insulation.
b. Winding data:
1. Insulation class.
2. Winding identification - same terminal designation as given in schematic and connection diagrams.
3. Wire size and insulation type, for example, AWG 10 type B2, etc.
4. Number of turns.
5. Nominal voltage between terminals.
6. Dc resistance at 25°C.
7. Material of insulation used between winding layers and between windings.
c. Varnish impregnation:
1. Varnish type and specification.
2. Varnish manufacturer and material identification.
3. Impregnation procedure - preheating temperature and time, number of dips and bakes, temperature and time of bakes, etc.
d. Volt-amp rating for potential transformers, current ratios for current transformers
(2) Semiconductor data including rating and complete ID for ordering from original supplier (JEDEC number is adequate), and the following:
a. Rectifiers using silicon diodes:
1. Circuit symbols and piece number.
2. Maximum reverse voltage peak and rms (complete rectifier circuit).
3. Maximum load current dc amperes (complete circuit).
4. Maximum diode case temperature at design ambient-use for diodes rated against case temperature.

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5. Maximum ambient temperature directly adjacent to diodes - use for diodes rated against ambient temperature - no cooling plates.
6. Cooling plate size, thickness and material where so mounted.
7. Circuit (for example, single phase bridge, etc.)
8. Total diodes in circuit.
9. Diodes in series per arm.
10. Diodes in parallel per arm.
11. Description of insulation hardware if insulated from a cooling plate.

b. Voltage regulator diodes:
1. Circuit symbol(s) and piece number.
2. Range of actual voltage levels which circuit will accept (with or without circuit adjustment) from regulator diodes and still give required equipment performance _____ volts to _____volts.
3. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance) plus or minus _____millivolts.
4. Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds):_____ma.
5. Maximum transient reverse current under any operating condition _____ma. for _____sec. duration and _____ frequency of occurrence.
6. Maximum ambient temperature at diode (no cooling plate).
7. Maximum case temperature of diode (complete for diodes rated against case temperature).
8. Cooling plate size, thickness and material where used.
9. Total diodes in circuit.
10. Diodes in series per circuit element (complete where more than one circuit element is used in the voltage regulator such as in a bridge type regulator circuit).
11. Description of insulation hardware if insulated from a cooling plate.

c. Voltage reference diodes (see regulator diodes also):
1. Circuit symbol(s) and piece number.
2. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s): ___volts to _____volts.
3. Error permitted in an installed voltage reference diode under all conditions of operation: plus or minus _____millivolts.
4. Circuit design current through voltage reference.
a. Nominal or average value: _____ma.
b. Regulation about nominal value: plus or minus _____ma.
5. Maximum ambient temperature (include internal heat rise above outside design ambient): _____a.
6. Number of voltage reference elements in series: (operation in parallel is not permitted).

d. Transistors:
1. Circuit symbol(s) and piece number.
2. Maximum collector to emitter voltage ____volts.
3. Maximum collector to base voltage _____volts.
4. Maximum collector current _____ma.
5. Maximum base current _____ma.
6. Maximum collector power dissipation _____mw.
7. Maximum ambient temperature (include internal heat rise at transistor at 50°C, design ambient _____C).
8. Maximum case temperature of transistor at 50°C, equipment design ambient and any condition of required equipment operation _____C. (Where rating is based on case temperature.)
9. Cooling plate size, thickness and material where so mounted.
10. Description of insulating hardware if insulated from a cooling plate.

e. Silicon controlled rectifiers:
1. Circuit symbol(s) and piece number.
2. Nominal working peak forward and reverse voltages.
3. Repetitive peak forward and reverse voltages (maximum).
4. Nonrepetitive peak forward and reverse voltages (maximum).

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5. False firing, or self-firing, restrictions.
6. Repetitive peak forward current (nominal and maximum values).
7. Surge current (peak for nonrepetitive value occurring within the conduction portion of a single cycle but after turn-on has occurred, effective amplitude and cycles duration for multi-cycle surges, or amplitude and duration for sub-cycle pulse duration).
8. Peak current and maximum rate of current rise during turn-on.
9. Minimum time from the instant of voltage reversal to the instant when voltage is re-applied in the forward direction.
10. Peak forward gate current (minimum and maximum, per device).
11. Peak forward gate voltage (minimum and maximum, per device).
12. Peak reverse gate voltage (maximum, per device).
13. Maximum gate power dissipation (average and peak, per device).
14. Maximum device case temperature at design ambient.
15. Cooling plate size, thickness and material, where so mounted.
16. Circuit.
17. Total devices in circuit.
18. Devices in series per arm.
19. Devices in parallel per arm.
20. Insulation hardware if insulated from a cooling plate.

f. Selenium surge suppressors:
   1. Identify the suppressor manufacturer's technical data bulletin applicable to each product certified for the application. For each application indicate which product is installed in the equipment and which is the alternate product that may be used for replacement purposes, when desired.
   2. Provide the following product rating data:
      a. Stack rms voltage (steady-state).
      b. Stack clamping voltages (new and aged).
      c. Peak discharge current (corresponding to clamping voltage).
      d. Polarization (polarized or nonpolarized).
      e. Maximum energy (watt-seconds) at rated clamping voltage (aged) for single pulse (nonrepetitive) current.
      f. Maximum pulse duration (microseconds) for repetitive pulses at supply frequency (specify cycles per second) and amplitude equal to clamping voltage (aged).
      g. Maximum ambient temperature at which rated values in 2.a. through e. apply.
      h. Maximum operating temperature.

g. Integrated circuits:
   1. The manufacturer and manufacturers type number shall be given.
   2. The function of the circuit shall be given along with a diagram illustrating the relationship of the output to the input.
   3. Maximum allowed currents and voltages shall be given for each terminal, as applicable.
   4. Maximum applied currents and voltages shall be given for each terminal, as applicable.
   5. Maximum allowed and applied power dissipation.
   6. Maximum ambient temperature.
   7. Maximum case temperature at 65°C, equipment design ambient, at worst case operating conditions.

(3) Resistor data including the type, resistance and wattage ratings, operating temperature and standard Military type number (if applicable). Actual power dissipated at design operating temperature.

(4) Capacitor data including the type, capacitance, voltage rating, maximum applied dc and ac voltage and frequency (values include transient), and standard Military type number (if applicable).

(5) Switch data including the type of switch, contact arrangement, number of of poles, number of positions, and voltage and current ratings of contacts.

(d) Electrical wiring and schematic diagrams:

(1) Schematic. A single electrical schematic diagram for each unit to represent clearly the operation of the equipment. The following features shall be incorporated in the diagram:

a. In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.

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b. A thin, broken line shall be used to represent the boundaries of each unit of subassembly. All terminals to which external connections are made shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.
c. Each part, such as resistors, capacitors, relays, shall be given a unique reference designation consisting of a letter denoting the type of part (as required by ANSI Y32.16-68) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of the electrical current or signal flow through the circuit.
d. In addition to the reference designation, the following information shall be noted adjacent to the part. Where numerical values are given, a code shall be noted to designate the units used:
1. Resistors. Resistance, power rating, and tolerance. If variable, an arrow to indicate clockwise rotation of the control shaft.
2. Capacitors. Capacitance, voltage rating, and tolerance.
3. Reactors. Inductance and current rating.
4. Semiconductors. Type number.
5. Integrated circuits. Type number, operational symbol, diagram of input/output relationship, terminal numbering corresponding to representative schematic.
6. Controls, adjustments and test points. The assigned name.
7. Voltage reference diodes. Type number and voltage.
e. All supply voltages, phases and frequencies and transformer terminal voltages shall be indicated and labeled as to purpose.
f. Winding polarities shall be indicated for transformers and saturable reactors. The same system shall be used throughout. The method of marking polarity shall be shown and explained in a note.
g. All schematic diagrams, size D and larger, shall be zoned. Zone widths shall be between 6 and 8 inches. Smaller size drawings may be zoned for clarity.
h. Description of operation. A brief description of operation of the circuit elements to enable authorities and operators to understand circuitry.

(2) Wiring diagram. A wiring diagram shall be drawn for each unit showing components and wiring in correct relation to physical positions. Wiring between parts and interconnection wiring shall be marked and color coded. Markings and part designations shall be consistent with other diagrams provided. Hook-up wire size and type shall be specified. Notes labeling terminals shall be provided for identification.

(e) Construction details:
(1) Two representative assembly views, as required, to show clearly the details of the design, construction and assembly of the unit and to identify each part and its location. Identification of parts shall correspond to the list of materials.
(2) Overall, drilling and mounting dimensions.
(3) Sectional views or notes as necessary to show or explain special mounting provisions or installation procedures (for example, mounting and module enclosure requirements for separately mounted modules).
(4) Details such as cable entrance provisions, gaskets, special mounting requirements and other details as applicable.
(5) Outline drawing of parts or components furnished separately (not mounted in an enclosure) to show critical dimensions including mounting dimensions.

(f) Installation drawings. Data to be shown on these drawings shall be as follows:
(1) Dimensional outline of all major assemblies (detector, module, etc.), showing overall and principle dimensions in sufficient detail to establish the limits of space in all directions required for installation, operation and servicing, exclusive of space required for personnel. Include the amount of clearances required for withdrawal of parts and subassemblies, clearances for cable and piping access, and clearances for ventilation. Also indicate the location of operating controls and test points.
(2) A table of reference drawings to include drawing number of each major assembly, weight of each enclosure and separately furnished assembly and heat dissipation of major assemblies.
(3) A block diagram of all major assemblies showing interconnection between these assemblies. Terminal identification, as well as maximum current rating for each cable should be shown.

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(4) Location, type and dimensions of cable entrance plates, connectors, with optional entrances indicated.
(5) Any special instructions for preservation, painting, installation or assembly as necessary.
(6) Heat dissipation.

(g) Drawing format. All of the above information may be presented on one sheet. However, separate sheets may be used for (c), (d), and (f), above. All other information, as required above, shall be presented on one sheet.

3.8 Technical manuals. Technical manuals shall be supplied in quantity as specified (see 6.2.1), and shall be in accordance with type I of MIL-M-15071 with content and arrangement as follows:

(a) Front matter.
(b) General information.
(c) Installation.
(1) This section is to include complete calibration and alignment procedures.
(d) Operation.
(e) Maintenance, repair and troubleshooting.
(f) Parts identification.
(g) Drawings - Reduced size drawing inserts.
(h) Appendix - Inserts on special components. One initial copy of the manual shall be supplied to NAVSEC for approval.

3.9 Onboard repair parts. The onboard repair parts shall be supplied in quantity as specified (see 6.2.1) and shall be in accordance with MIL-P-15137 (see 6.3).

3.10 Equipment performance. Performance tolerances apply to both sensing techniques unless stated otherwise, and are expressed in percent of equipment span unless stated otherwise.

3.10.1 Static error band. The static error band shall not exceed plus or minus 1 percent (see 4.8.1).

3.10.2 Repeatability. Repeatability of the output shall not exceed 0.3 percent (see 4.8.1).

3.10.3 Threshold and deadband. The maximum least detectable increment plus deadband shall not exceed 0.2 percent (see 4.8.3).

3.10.4 Ripple. Equipment output ripple shall not exceed 0.15 percent of full scale output (see 4.8.4).

3.10.5 Warm-up time. The equipment output shall attain a value within plus or minus 1 percent of the steady state output with no overshoot in excess of 1 percent. Output shall reach this band in 30 minutes or less and shall remain in this band (see 4.8.5).

3.10.6 Inclination. Maximum deviation of equipment output resulting from inclination shall not exceed 3.0 percent (see 4.8.6).

3.10.7 Input resistance. The equipment output shall remain within the static error band (see 3.10.1 and 4.8.7).

3.10.8 Supply voltage and frequency (steady state). Maximum difference between outputs at any voltage and frequency condition and the normal (115 volts, 60 Hz) at the same input shall not exceed 1.0 percent (see 4.8.8).

3.10.9 Temperature error. Temperature error based on the test temperature extremes of the equipment shall not exceed 0.04 percent per degree Fahrenheit (see 4.8.15).

3.10.10 Response time. The 95 percent response time shall be 0.5 seconds or less (see 4.8.9).

3.10.11 Supply voltage and frequency (transients) (see 4.8.10).

3.10.11.1 Voltage. During the voltage transient test the equipment output shall remain within plus or minus 5 percent of the steady state output.

3.10.11.2 Frequency. During the frequency transient test the equipment output shall remain within plus or minus 0.5 percent of the steady state output.

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3.10.12 Temperature. Performance shall be within the static error band specified in 3.10.1 (see 4.8.11).

3.10.13 Insulation resistance. The insulation resistance of the equipment between circuits and between circuits and ground shall be not less than 10 megohms (see 4.8.12).

3.10.14 Vibration test. Monitored output during all phases of vibration test shall show no variation from steady state output in excess of 2.0 percent. There shall be no visible evidence of damage to the equipment as a result of the vibration test (see 4.8.13).

3.10.15 Shock test. After the shock test but prior to any adjustment the equipment output shall show no deviation greater than 3 percent. A post test calibration, using adjustments provided, shall conform to performance requirements of 3.10.1. There shall be no visual evidence of damage to the equipment as a result of the shock test (see 4.8.14).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.2.1 and 6.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

(a) Qualification tests (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Periodic tests (see 4.5).

4.3 Qualification tests.[1/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.3.1.

4.3.1 Samples for qualification. The number of samples to be submitted for the qualification tests specified in table I shall depend on the equipment design. If each range is covered by a separate and distinct design, a sample for each range will require testing. Only one sample of a basically similar design series which may cover many ranges need be submitted for testing if mechanical and electrical similarity is deemed sufficient by NAVSLC. The range which has successfully passed the qualification tests for this design will qualify all other ranges of the same design.

Table I - Qualification inspection.

Examination and tests	Requirement paragraph	Test paragraph
General examination	3.4	4.6
Output	3.4.2.2	4.8.1
Static error band	3.10.1	4.8.1
Repeatability	3.10.2	4.8.1
Threshold and deadband	3.10.3	4.8.3
Ripple	3.10.4	4.8.4
Warm-up time	3.10.5	4.8.5
Inclination	3.10.6	4.8.6
Input resistance	3.10.7	4.8.7
Supply voltage and frequency (steady state)	3.10.8	4.8.8

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4 and 6.4.1).

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Table I - Qualification inspection (cont'd).

Examination and tests	Requirement paragraph	Test paragraph
Temperature error	3.10.9	4.8.15
Response time	3.10.10	4.8.9
Supply voltage and frequency (transient)	3.10.11	4.8.10
Temperature	3.10.12	4.8.11
Insulation resistance	3.10.13	4.8.12
Vibration	3.10.14	4.8.13
Shock	3.10.15	4.8.14

4.4 Quality conformance inspection. All temperature measurement equipment offered for delivery shall be subjected to the examination and tests listed in table II and shall be conducted in the order listed. Failure of any temperature equipment to meet the requirements of this specification shall be cause for rejection.

Table II - Quality conformance inspection.

Examination and tests	Requirement paragraph	Test paragraph
General examination	3.4	4.6
Output	3.4.2.2	4.8.1
Static error band and repeatability	3.10.1 and 3.10.2	4.8.1
Ripple	3.10.4	4.8.4
Input resistance (thermocouple sensing technique only)	3.10.7	4.8.7
Insulation resistance	3.10.13	4.8.12

4.4.1 Quality conformance inspection reports. A test report shall be packed with each unit to attest to the performance of the examination and tests specified in 4.4. Reports shall include a complete identification of the equipment with a check notation of "passed" or "not passed" for each examination and test, the requirement paragraph and applicable test paragraph. A calibration record shall be included to determine conformance with the requirements of 3.10.1 and 3.10.2.

4.5 Periodic tests. Three samples shall be selected at random out of every 100 units which have satisfactorily passed the tests in 4.4 from each lot offered for delivery and shall be subjected to the tests in table III. If any sample fails in any test, no deliveries shall be accepted for quality conformance inspection until the manufacturer has determined the cause of the defect and has taken the necessary action to correct or eliminate the defect from all units on hand. The failed test and any other periodic test required by the Government shall be repeated to demonstrate that the corrective action will enable the equipment to conform to the requirements of this specification.

Table III - Periodic tests.

Tests	Requirement paragraph	Test paragraph
Input resistance	3.10.7	4.8.7
Supply voltage and frequency (steady state)	3.10.8	4.8.8
Temperature error	3.10.9	4.8.15
Response time	3.10.10	4.8.9
Temperature	3.10.12	4.8.11
Insulation resistance	3.10.13	4.8.12

4.5.1 Periodic test reports. A complete report on all periodic tests conducted as specified in 4.5 shall be submitted to NAVSEC for review. The format and content of the report shall conform to MIL-STD-831.

4.6 General examination. The completed equipment shall be given a thorough examination to determine that it conforms to this specification and applicable drawings with respect to material, finish, workmanship, construction, assembly, dimensions, weight, marking of identification and information plates. This examination shall be limited to those examinations

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that may be performed without disassembling the unit in such a manner that its performance, durability or appearance would be affected. This examination shall include a check of all operating controls and adjustments, as applicable (see 3.4, 3.4.2.5, and 3.4.2.6).

4.7 Test conditions. Except where the following factors are the variables, the tests specified in 4.8 shall be conducted with the equipment operating under the following conditions:

(a) Ambient temperature shall be 25° ± 5°C nominal.
(b) Relative humidity shall be ambient nominal.
(c) Supply voltage shall be 115 volts nominal.
(d) Supply frequency shall be 60 Hz nominal.
(e) Control positions shall be neutral or normal.
(f) Specified external load (see 3.4.2.2).

4.8 Tests.

4.8.1 Static error band and repeatability. Prior to the start of this test, the test sample and associated test equipment shall be energized for a period of two hours. The test sample shall first be cycled over its full temperature range by slowly increasing and decreasing the input for six continuous cycles. The calibration measurements shall be made at a minimum of 5 equally spaced intervals over the entire range (both upscale and downscale). Precaution shall be taken to avoid overshoot. This calibration procedure shall be applied three successive times to determine repeatability. Static error band of all calibrations shall meet the requirements of 3.10.1. Repeatability shall meet the requirements of 3.10.2.

4.8.2 Reference measurement. The following reference measurement shall be conducted when specified in the individual test:

(a) A one trial calibration with at least five equally spaced intervals over the entire input range both upscale and downscale.

4.8.3 Determination of threshold and deadband. Threshold and deadband shall be determined with 80 ± 5 percent of the temperature span applied to the test sample. The temperature interval beyond this point required to produce a detectable change in output and the temperature interval in the opposite direction to return the output to the original value shall be determined in each direction. Threshold is the first temperature interval. The second temperature interval is the sum of the threshold and deadband. Performance shall conform to the requirements of 3.10.3.

4.8.4 Ripple-equipment output. Ripple shall be determined at an input of 80 ± 5 percent of the temperature span. Performance shall conform to the requirements of 3.10.4.

4.8.5 Determination of warm-up time. This test shall be conducted to determine the elapsed time between the application of line power to the test sample and the point at which the equipment output reaches the conditions specified in 3.10.5.

4.8.5.1 The test sample shall be placed in an ambient temperature of 25 ± 2°C for not less than 2 hours de-energized. Recording equipment and other auxiliary equipment shall be energized to assure complete warm-up. An input of 80 ± 5 percent of the span is to be applied to the test sample and maintained constant during this test. Performance shall conform to 3.10.5.

4.8.6 Inclination. The equipment shall be inclined for a period of at least 1 minute in each of the following positions:

(a) 45 degrees forward.
(b) 45 degrees backward.
(c) 45 degrees to the left.
(d) 45 degrees to the right.

In each position a reference measurement (see 4.8.2) shall be made. Performance shall conform to 3.10.6.

4.8.7 Input resistance. For thermocouple sensing technique, a noninductive resistance of 150 ± 1 percent ohms shall be placed in each lead to the input terminals to total a circuit resistance of approximately 300 ohms. A reference measurement (see 4.8.2) shall be made. Performance shall conform to 3.10.7.

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4.8.8 Supply voltage and frequency (steady state). The equipment shall be operated at the various combinations of normal, maximum and minimum steady state voltages and frequencies outlined in 3.4.2.11. The equipment shall be operated at least 15 minutes in each combination during which time a reference measurement (see 4.8.2) shall be taken. The above shall be performed at ambient temperatures of 0° ± 2°C, 25° ± 2°C, and 65° ± 2°C. Performance shall conform to 3.10.8.

4.8.9 Response time. A step input equal to 10 to 90 percent of full scale equipment temperature span shall be applied to the test sample. Performance shall conform to 3.10.10.

4.8.10 Supply voltage and frequency (transient). The tests specified in 4.8.10.1 through 4.8.10.2 shall be conducted with a temperature input signal equal to 80 ± 5 percent of the input span. Performance shall conform to 3.10.11.

4.8.10.1 Transient voltage.

4.8.10.1.1 With the equipment operating on the upper limit of steady voltage, a transient voltage of plus 20 percent of nominal voltage recovering to the steady state band in 2 seconds shall be superimposed.

4.8.10.1.2 With the equipment operating on the lower limit of steady state voltage, transient voltage of minus 20 percent of nominal voltage recovering to the steady state band in 2 seconds shall be superimposed.

4.8.10.2 Transient frequency. With the equipment operating at the nominal frequency, a transient frequency of 2 cycles per second shall be applied of which not more than 1/2 cycle per second is outside the steady state tolerance band recovering to the steady state tolerance band within 2 seconds. This shall be repeated with a minus 2 cycles per second transient frequency.

4.8.11 Temperature test. The equipment shall be capable of normal operation (without alignment or adjustment, other than the accessible controls employed for operation of the equipment) throughout the following temperature cycle; tolerances in operating characteristics shall be as specified in the individual equipment specification:

(a) Hold room temperature at 0° ± 2°C for at least 24 hours.
(b) Increase room temperature in steps of 10 degrees each, at 30 minutes per step, until + 65° ± 2°C is reached, and hold at that temperature for at least 4 hours.
(c) Reduce room temperature in steps of 10° each, at 30 minutes per step, until + 25° ± 2°C is reached, and hold at that temperature for at least 4 hours. At each temperature plateau (0°C, 65°C and 25°C), a reference measurement (see 4.8.2) shall be made. Performance shall conform to 3.10.12. Temperature error shall be as specified in 3.10.9.

4.8.12 Insulation resistance. The insulation resistance of the equipment shall be determined by applying 50 volts dc between electrical input and output circuits and between these circuits and ground (see 3.10.13).

4.8.13 Vibration. The equipment shall be tested in accordance with type I (environmental vibration) of MIL-STD-167 except that the upper limit of the frequency range shall be 100 Hz. The amplitudes of vibration shall be in accordance with table IV. If no resonances are observed, the two hour endurance test shall be conducted at 100 Hz. During the vibration test, an input equal to 80 ± 5 percent of the temperature span shall be applied to the test sample. The output during the test shall be monitored. Performance shall conform to 3.10.14.

Table IV - Amplitudes of vibration.

Frequency range	Table amplitude
Hz	Inch
5 to 20	0.030 ± 0.006
21 to 50	0.020 ± 0.004
51 to 100	0.010 ± 0.002

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4.8.14 Shock. The shock test shall be conducted in accordance with MIL-S-901, grade A, class I, type C. During the test an input equal to 80 ± 5 percent of the temperature span shall be applied to the equipment. The test sample output during the test shall be monitored. Before and after this test, reference measurements shall be made for comparison (see 4.8.2). Performance shall conform to 3.10.15.

4.8.15 Temperature error. The temperature error shall be determined from the data obtained in 4.8.8 at nominal voltage and frequency conditions. Performance shall meet the requirements of 3.10.9.

4.9 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning. Unless otherwise specified herein, prior to packaging, the equipment shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Packaging.

5.1.1.2.1 Temperature measurement equipment. Equipment shall be individually cushioned and packaged in accordance with method III of MIL-P-116 one to a box (see 5.1.1.2.2).

5.1.1.2.2 Unit containers. Unit containers shall conform to any of the following containers at the option of the supplier:

Specification	Classification
PPP-B-566	Variety 4, style 1
PPP-B-636	Type CF, weather resistant
PPP-B-665	Style A
PPP-B-676	Type 1, class A

Container closure and sealing shall be in accordance with the applicable container specification or appendix thereto.

5.1.2 Level C. Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage. This level may conform to the supplier's commercial practice.

5.2 Packing.

5.2.1 Level A. Equipment, packaged as specified in 5.1.1.2, shall be packed in weather resistant fiberboard containers in accordance with PPP-B-636 or class 2, fiberboard-corrugated triple-wall containers in accordance with PPP-B-640, at the supplier's option. All center and edge seams and manufacturer's joints shall be sealed and waterproofed with pressure sensitive tape in accordance with the applicable fiberboard box specification or appendix thereto. Shipping containers shall be closed and banded in accordance with the applicable box specification or appendix thereto. Banding (reinforced) shall be accomplished with use of pressure sensitive, reinforced tape. The gross weight of triplewall boxes should not exceed 250 pounds. PPP-B-636 containers shall not exceed the weight limitations of the specifications. When specified, unit fiberboard containers conforming to weather resistant class of PPP-B-636 closed, sealed and banded as specified herein may be used as the shipping container and need not be overpacked.

5.2.2 Level B. Equipment, packaged as specified in 5.1.1.2, shall be packed in containers conforming to PPP-B-636, domestic type or PPP-B-640, class 1. Shipping containers shall be closed in accordance with the applicable container specification or appendix thereto, except that fiberboard containers may be sealed with PPP-T-45 tape. Shipping containers shall not exceed the weight limitations of the applicable box specification. Unit fiberboard containers conforming to PPP-B-636 closed as specified herein may be used as the shipping container and need not be overpacked.

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5.2.3 Marking. In addition to any special marking requirements, interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

5.2.4 Repair parts (submarine, onboard and stock). Repair parts shall be preserved and packaged level A; packed level A, B, or C and marked in accordance with MIL-E-17555. Level A preservation and packaging methods shall be modified in accordance with MIL-STD-758 for items that can be stored in bins or drawers.

6. NOTES

6.1 Intended use. This equipment is intended to provide control input and monitoring of temperatures for shipboard engineering plants.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Classification required (see 1.2.1).
(c) Quantity of units required.
(d) Type of enclosure mounting (see 3.4.2.9).
(e) Power requirements (see 3.4.2.11).
(f) Equipment temperature ranges (see 3.5.1.1 and 3.5.2.1), or as specified.
(g) Type and quantity of drawings required (see 3.7.2).
(h) Quantity of manuals required (see 3.8).
(i) Onboard repair parts required (see 3.9).
(j) Quality assurance requirements (see 4.1.1).
(k) Level of preservation, packaging, packing and marking required (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Drawings (see 3.7).
(b) Technical manuals (see 3.8).
(c) Quality conformance inspection reports (see 4.4.1).
(d) Periodic test reports (see 4.5.1).

6.3 Management control system documents. The following management control system documents should be included on DD Form 1660:

(a) MIL-P-15137 (see 3.9).
(b) MIL-I-45208 (see 4.1.1).

6.4 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 24387 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, Center Building, Prince George's Center Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4.1).

6.4.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.6 THE MARGINS OF THIS SPECIFICATION ARE MARKED "*" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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